Diffusion Models for Discrete Data

CS236: Deep Generative Models — Lecture 18

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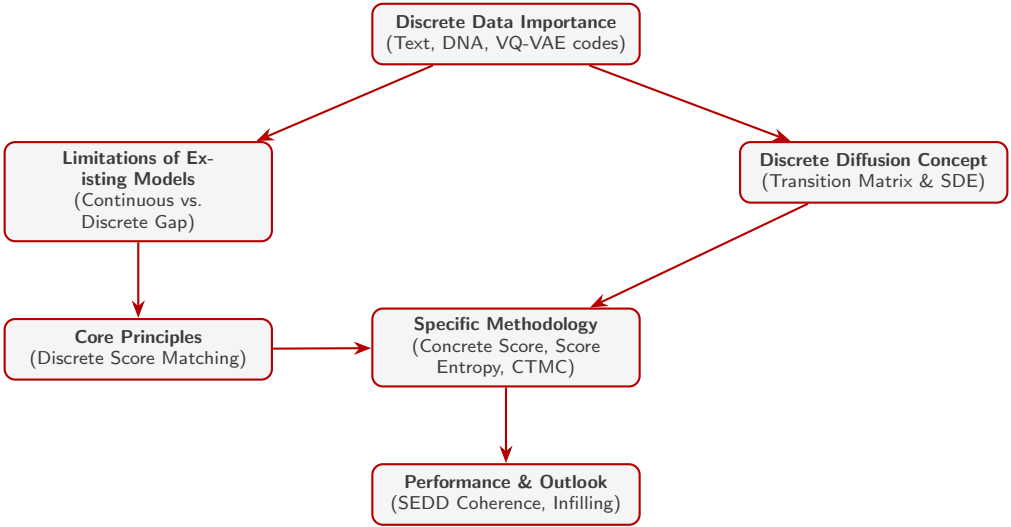
1.1. Introduction

Background and Motivation

- **Success in Continuous Data**: Generative models (GANs, VAEs, Continuous Diffusion) have achieved spectacular success in continuous domains like image and audio.
- **Challenges in Discrete Domains**: Innate structures of discrete data—such as natural language text, DNA sequences, and chemical structures—pose persistent challenges for gradient-based continuous models.
- **Objective**: This presentation explores the core mechanics, learning objectives (*Score Entropy*), and performance of **Discrete Space Diffusion Models** as a mathematically rigorous alternative to continuous approximations.

1.2. Presentation Roadmap



2.1. Why Discrete Data Matters

Discrete data is highly ubiquitous and represents the structural basis of various crucial domains.

Language Data (Text)

- Formed as a non-continuous sequence of tokens.
- Pre-training LLMs is fundamentally about learning massive **discrete probability models**.

Biological Sequences

- DNA sequences and protein amino acids are inherently discrete.
- Accurate generative modeling opens new horizons in drug discovery.

Discretized Images

- Modern frameworks (VQ-VAE, MAGVIT-v2) quantize continuous pixel values into discrete latent spaces, showing improved generative efficiency.

2.2. Mathematical Bottlenecks of Generation

Traditional generative frameworks are designed for continuous spaces $\mathbb{R}^d$ and face critical failures when directly applied to discrete spaces $\{1, \ldots, N\}^d$.

- **Calculus Dependency**:
  - **GANs and Flow-based Models** rely heavily on backpropagation gradients and variables transformation (Jacobians).
  - Restricting outputs to discrete space cuts off the gradient path, making conventional training mathematically impossible.
- **Failure of Embedding & Rounding**:
  - A naive strategy: embed tokens into $\mathbb{R}^d$, apply continuous diffusion, and round back to the nearest token during sampling.
  - **Problem**: Token embedding spaces are extremely high-dimensional and sparse. The continuous model spends vast compute exploring empty space, resulting in poor sample quality and extremely slow generation (often requiring $\sim$4,000 steps).

3.1. Autoregressive (AR) Modeling Mechanics

Sequential Factorization

AR decomposes the joint probability of a sequence $X = (x_1, \ldots, x_D)$ into causal conditional factors:

$$P(X) = \prod_{i=1}^{D} P(x_i \mid x_1, \ldots, x_{i-1})$$

- Mimics the human process of reading and writing sequentially.
- Integrates flawlessly with **Transformer** architectures utilizing causal attention masking.

Key Strengths

- **Scalability**: Computes the probability distribution over N tokens for only one step at a time.
- **Theoretical Capacity**: Given sufficient capacity, AR models can theoretically approximate any joint sequence distribution.
- **Natural Inductive Bias**: Pairs naturally with sequential text structures.

3.2. Structural Weaknesses of AR Models

Despite their massive success in NLP, AR models exhibit severe bottlenecks:

- **Sampling Drift (Error Accumulation)**:
  - During generation, minor errors at early steps propagate downstream. As the sequence length increases, the model drifts away from the true data distribution.
- **Acausal / Non-sequential Data Mismatch**:
  - For biological sequences (e.g., DNA, proteins) or physical networks, there is no natural "left-to-right" causality, making AR's inductive bias forced and unnatural.
- **Sequential Generation Latency**:
  - Must generate tokens sequentially (Token-by-Token). This sequential bottleneck cannot be parallelized, leading to major inference speed constraints.

4.1. Generalizing Score Matching: Concrete Score

Since the continuous score $\nabla_x \log p(x)$ is undefined in discrete domains, we use finite differences to establish the relative probability ratio.

Mathematical Definition of Concrete Score

For a state x and any neighboring state y, the score is defined as:

$$\text{Concrete Score} \equiv \left[\frac{p(y)}{p(x)}\right]_y$$

- **Local Neighbor Restructuring for Computational Feasibility**:
  - Computing this ratio over all arbitrary states scales poorly at $\mathcal{O}(N^{2d})$.
  - **Solution**: Restrict y to *local neighbors* differing by exactly one token. This simplifies the complexity to $\mathcal{O}(N \cdot d)$ and allows modeling via standard Seq-to-Seq networks.

4.2. Learning the Concrete Score: Score Entropy

Fundamental Score Entropy Loss

A loss function that forces a neural network $s_\theta(x)_y$ to accurately estimate the probability ratio $\frac{p(y)}{p(x)}$ without calculating the partition function:

$$\min_\theta \mathbb{E}_{x\sim p} \sum_{y\neq x} \left(s_\theta(x)_y - \frac{p(y)}{p(x)} \log s_\theta(x)_y \right)$$

- **Challenge**: The true data distribution $p(x)$ is unknown, making the ratio $\frac{p(y)}{p(x)}$ incomputable.
- **Solution: Denoising Score Entropy (DSE)**:
  - By corrupting the clean data $x_0 \sim p_0$ with a transition kernel $p_t(x|x_0)$, we bypass $p(x)$ and derive a computable conditional objective:

$$\mathrm{DSE}(\theta) = \mathbb{E}_{x_0\sim p_0, x\sim p_t(\cdot|x_0)} \sum_{y\neq x} \left(s_\theta(x,t)_y - \frac{p_t(y|x_0)}{p_t(x|x_0)} \log s_\theta(x,t)_y \right)$$

4.3. Discrete Diffusion via Continuous Time Markov Chains

Instead of discrete SDEs, the forward noise injection is modeled as a Continuous Time Markov Chain (CTMC) over the probability vector p_t.

Forward Transition Equation

$$\frac{dp_t}{dt} = Q_t p_t$$

Where the Transition Rate Matrix Q_t satisfies:

1. Columns sum to 0 ($\sum_j Q_t(j,i) = 0$)
2. Non-diagonal entries are non-negative ($Q_t(j,i) \geq 0, \quad j \neq i$)

- **Absorbing (Masking) Transition**: Tokens gradually transition to a designated [MASK] token.
- **Uniform Transition**: Tokens jump randomly to any other token in the vocabulary.

4.4. Reverse Diffusion & Sampling Acceleration

Sampling is performed by running the Markov chain backward in time using the learned Concrete Scores.

Reverse Transition Rate Matrix $\bar{Q}_t$

$$\bar{Q}_t(j,i) = \frac{p_t(j)}{p_t(i)} Q_t(i,j) \approx s_\theta(i,t)_j Q_t(i,j)$$

- **Discretization for High Parallelism**:
  - Instead of updating one token at a time, we group multiple reverse infinitesimal transitions.
  - For instance, we can unmask or flip hundreds of tokens simultaneously.
  - This reduces the total required steps to a mere 64–128 steps, enabling extremely **fast parallel sampling**.

5.1. Coherent Generation and Speed Trade-off

The prominent discrete diffusion model—**SEDD** (Score Entropy Discrete Diffusion)—displays powerful capabilities on text generation.

- **Coherence vs. GPT-2**:
  - Outperforms or closely matches GPT-2 in generating long, semantically coherent text sequences under standard datasets (e.g., Wikitext).
- **Flexible Compute-Quality Trade-off**:
  - **Fast Sampling (64 steps)**: Generates text highly rapidly, retaining comparable quality to sequential autoregressive models.
  - **High-Quality Sampling (248+ steps)**: Shows a log-linear perplexity improvement as steps scale up, yielding high-fidelity text unmatched by fixed-compute models.

5.2. Controllable Generation via Prompt Infilling

Unlike autoregressive models that are strictly unidirectional, discrete diffusion is natively bidirectional and allows for flexible infilling.

Prompt Infilling Mechanics

- **Bidirectional Contextualization**: Given a prefix and a suffix (or arbitrary fragments), the model fills in the missing [MASK] regions globally and simultaneously.
- **Guided Generation**: Enables conditional steering (e.g., maintaining specific motifs in DNA sequences or satisfying exact structural constraints in natural language).

5.3. Rigorous Likelihood Evaluation

Mathematical Likelihood Bound

- The Denoising Score Entropy loss mathematically acts as a **likelihood upper bound** for the true sequence perplexity (PPL).
- SEDD achieves competitive or superior Perplexity compared to GPT-2 across multiple benchmark datasets, proving its strength as a density estimator.

Model	LAMBADA	WikiText2	PTB	WikiText103	1BW
GPT-2-small	**45.04**	**42.43**	138.43	**41.60**	**75.20**
SEDD-small Absorb	$\leq$ 52.21	$\leq$ 44.75	$\leq$ **130.49**	$\leq$ 43.14	$\leq$ 80.70

Table: Likelihood / Perplexity (PPL) Upper Bounds comparison: GPT-2 vs. SEDD

6.1. Core Takeaways of Discrete Diffusion

Paradigm Shift

- **Calculus Independence**: Liberates discrete generation from continuous constraints (rounding, Jacobian calculation, and continuous embedding exploration).
- **Rigorous Formulation**: Concrete Score and DSE provide a robust, formal, and mathematically sound probability foundation.

Practical Value

- **No Sampling Drift**: Multi-step simultaneous denoising naturally diminishes the error accumulation found in causal sequential models.
- **Parallel Execution**: Achieves fast parallel decoding that utilizes modern GPU hardware efficiently.

6.2. Future Directions

Discrete space diffusion opens exciting paths across multiple frontiers:

- **Scaling to LLMs**:
  - Serving as a fast parallel-decoding alternative to standard autoregressive transformers.
- **Revolution in Bioinformatics**:
  - Tailor-made for biologically acausal sequences like DNA, RNA, and protein folds.
- **Unified Multimodal Architectures**:
  - Synthesizing quantized vision codes (VQ-VAE) and language tokens into a unified, mathematically consistent discrete diffusion pipeline.

Thank You